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MINISTRY OF EDUCATION, SINGAPORE
in collaboration with
CAMBRIDGE ASSESSMENT INTERNATIONAL EDUCATION
General Certificate of Education Normal (Academic) Level

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MATHEMATICS (SYLLABUS A) 4045/02

Paper 2

September/October 2025

2 hours

Candidates answer on the Question Paper.

READ THESE INSTRUCTIONS FIRST

Write your Centre number, index number and name on all the work you hand in.
Write in dark blue or black pen.
You may use an HB pencil for any diagrams or graphs.
Do not use staples, paper clips, glue or correction fluid.
DO **NOT** WRITE ON ANY BARCODES.

Section A
Answer **all** questions.

Section B
Answer **one** question.

The number of marks is given in brackets [ ] at the end of each question or part question.

If working is needed for any question it must be shown with the answer.
Omission of essential working will result in loss of marks.
The total number of marks for this paper is 70.

The use of an approved scientific calculator is expected, where appropriate.
If the degree of accuracy is not specified in the question and if the answer is not exact, give the answer to three significant figures. Give answers in degrees to one decimal place.
For π, use either your calculator value or 3.142.

This document consists of **19** printed pages and **1** blank page.

Singapore Examinations and Assessment Board

Cambridge Assessment International Education

Mathematical Formulae

Compound interest

$$\text{Total amount} = P\left(1+\frac{r}{100}\right)^n$$

Mensuration

$$\text{Curved surface area of a cone} = \pi r l$$

$$\text{Surface area of a sphere} = 4\pi r^2$$

$$\text{Volume of a cone} = \frac{1}{3}\pi r^2 h$$

$$\text{Volume of a sphere} = \frac{4}{3}\pi r^3$$

$$\text{Area of triangle } ABC = \frac{1}{2}ab\sin C$$

$$\text{Arc length} = r\theta, \text{ where } \theta \text{ is in radians}$$

$$\text{Sector area} = \frac{1}{2}r^2\theta, \text{ where } \theta \text{ is in radians}$$

Trigonometry

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

$$a^2 = b^2 + c^2 - 2bc\cos A$$

Statistics

$$\text{Mean} = \frac{\Sigma fx}{\Sigma f}$$

$$\text{Standard deviation} = \sqrt{\frac{\Sigma fx^2}{\Sigma f} - \left(\frac{\Sigma fx}{\Sigma f}\right)^2}$$

Section A (62 marks)

Answer **all** the questions in this section.

1 Ashraf records the speeds of 50 vehicles passing his apartment.
He shows these on this stem-and-leaf diagram.

3	5	7	8	8	9							
4	2	3	3	5	6	8	9	9				
5	1	2	2	4	5	6	7	7	8	8	9	9
6	0	3	3	6	6	6	7	8	9	9		
7	0	0	1	2	3	5	6	6	8			
8	1	1	3	4	6	7						

Key: 3 | 5 = 35 km/h

(a) Find the modal speed.

Answer .... km/h [1]

(b) Find the median speed.

Answer .... km/h [1]

(c) One of the 50 vehicles is chosen at random.

Find the probability that the vehicle was travelling at less than 50 km/h.

Answer .... [1]

2 Devi and Kang each have \$500 to invest.

(a) Devi invests \$500 in an account that pays 2.3% per year compound interest.

Calculate the amount in the account at the end of 6 years.

Answer \$ .... [2]

(b) Kang invests \$500 in an account that pays simple interest at 2.8% per year.

Find the difference between the amounts in Devi's account and Kang's account at the end of 6 years.

Answer \$ .... [3]

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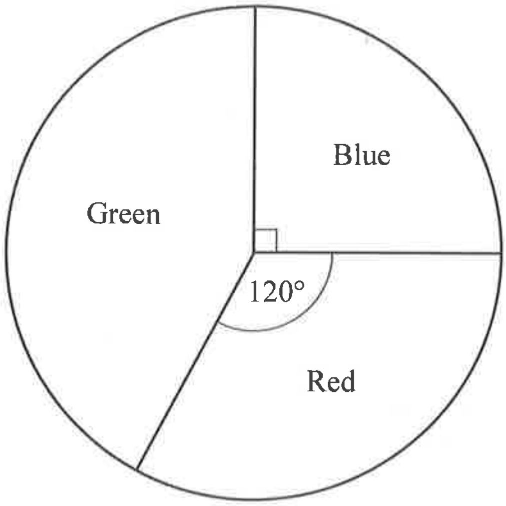
3



The pie chart represents the numbers of votes received by three parties in an election.
4200 people voted for the Blue Party.

(a) Find the total number of people who voted.

Answer .... [1]

(b) Find the number of people who voted for the Green Party.

Answer .... [3]

4 **(a)** By writing each number correct to 1 significant figure, estimate the value of

$$\frac{4.936 \times \sqrt{9.7 - 0.96}}{5.765}.$$

Answer .... [2]

(b) Write 20.7 million in standard form.

Answer .... [1]

(c) $P = a \times 10^m$ where $4 < a < 10$ and m is an integer.
$Q = b \times 10^n$ where $4 < b < 10$ and n is an integer.

Written in standard form, $PQ = c \times 10^r$, where $1 \leqslant c < 10$.

(i) Lee says that $c = a \times b$.

Explain why Lee is wrong.

Answer ....

.... [1]

(ii) Find c in terms of a and b.

Answer $c =$ .... [1]

(iii) Find r in terms of m and n.

Answer $r =$ .... [1]

5 **(a)** Maya, Nadia and Rahim share \$300 in the ratio 4 : 5 : 3.

How much do they each receive?

Answer Maya \$ ....

Nadia \$ ....

Rahim \$ .... [3]

(b) A map is drawn to a scale of 1 : *n*.
A forest has an area of $2\,km^2$.
The area of the forest on the map is $8\,cm^2$.

Find the value of *n*.

Answer $n =$ .... [2]

6 **(a)** Uma drives from P to Q at an average speed of x km/h.
It takes her 1 hour 30 mins.

She then drives from Q to R at an average speed of $(x-8)$ km/h.
This part of the journey takes her 1 hour 15 mins.

The total distance Uma travels is 111 km.

Write down an equation in x and solve it to find x.

Answer $x =$ .... [3]

(b) Factorise $6x^2 - 7x - 10$.

Answer .... [2]

(c) Factorise completely $ax - 2bx + 3ay - 6by$.

Answer .... [2]

(d) Solve $3x^2 - 7x - 5 = 0$.
Give your answers correct to 2 decimal places.

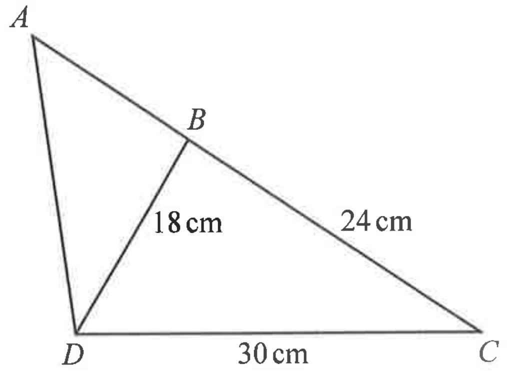
Answer $x =$ .... or $x =$ .... [3]

7 **(a)** The interior angle of a regular polygon is 156°.

Find the number of sides of the polygon.

Answer .... [2]

(b)



$BC = 24\,\text{cm}$, $BD = 18\,\text{cm}$ and $DC = 30\,\text{cm}$.
ABC is a straight line.

(i) Show that angle DBC is a right angle.

Answer

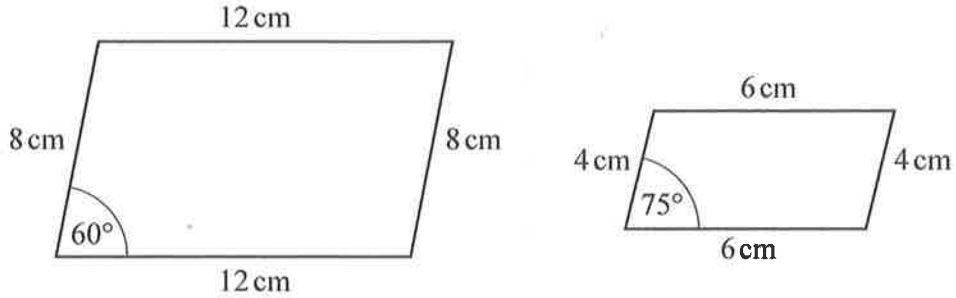
[2]

(ii) The area of triangle $ADC = 337.5\,\text{cm}^2$.

Find the length of AB.

Answer .... cm [3]

(c)

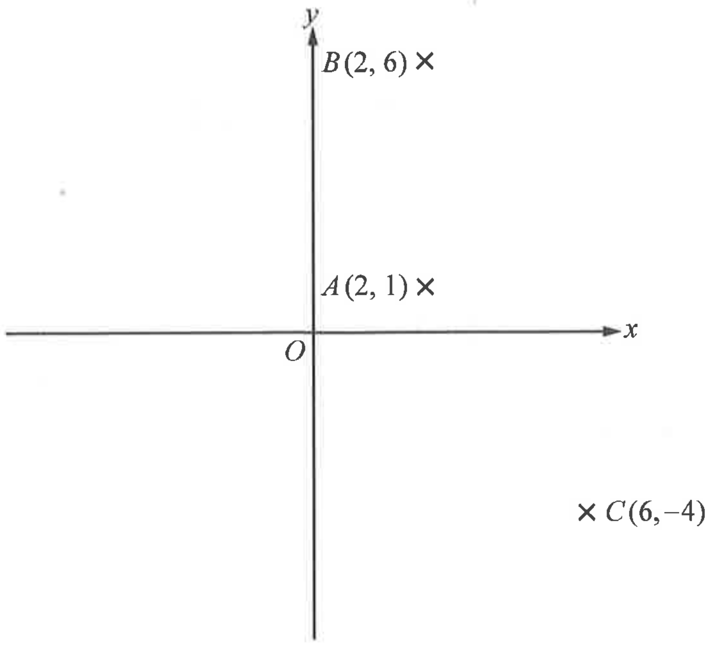


Meihui says that the sides of the parallelograms are proportional and so the parallelograms are similar.

Explain why Meihui is **not** correct.

Answer .... [1]

8 A is the point (2, 1), B is the point (2, 6) and C is the point (6, −4).



(a) Find the equation of line AB.

Answer .... [1]

(b) Find the equation of line BC.

Answer .... [3]

(c) Find the area of triangle ABC.

Answer .... square units [3]

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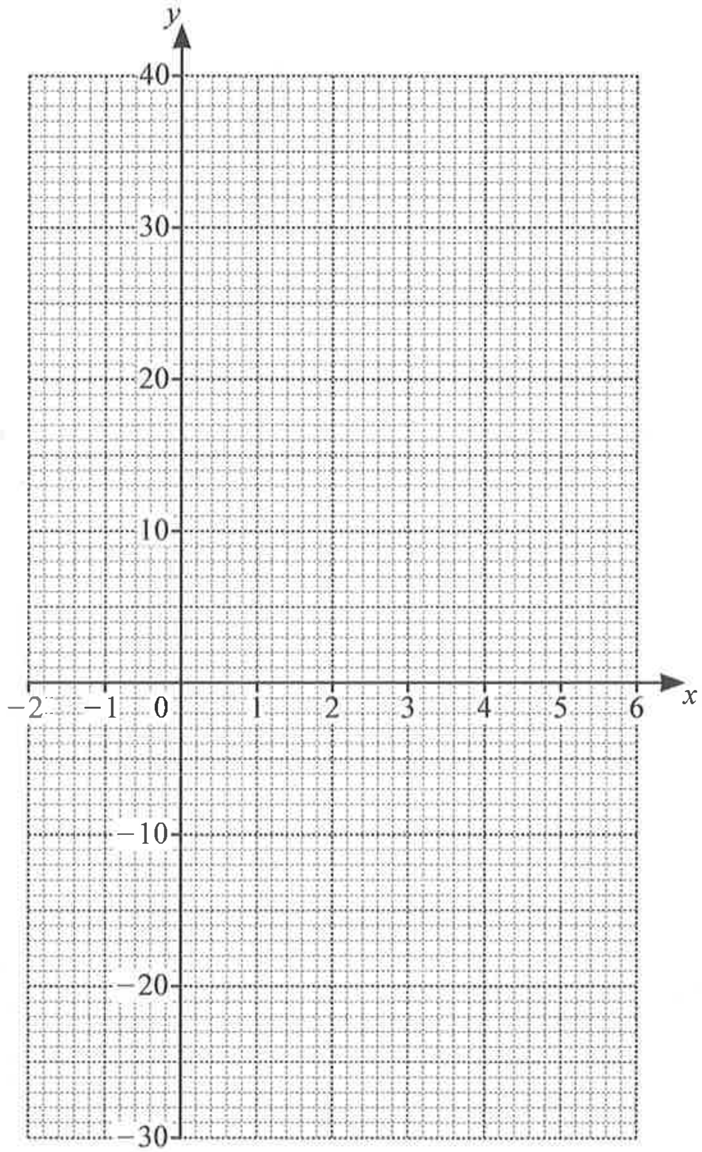
9 **(a)** Complete the table of values for $y = x^3 - 5x^2 + 4$.

x	–2	–1	0	1	2	3	4	5	6
y	–24		4	0	–8	–14	–12		40

[2]

(b) Draw the graph of $y = x^3 - 5x^2 + 4$ for $-2 \leqslant x \leqslant 6$.

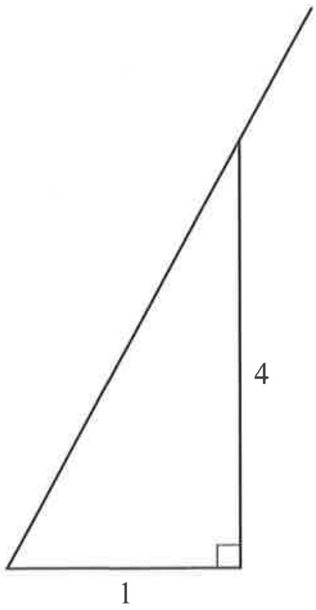


[3]

(c) Use your graph to estimate the value of x when $y = 20$.

Answer .... [1]

10 To use a ladder safely, the 4 to 1 rule should be used.
The rule is that the vertical height should be 4 times the horizontal distance.



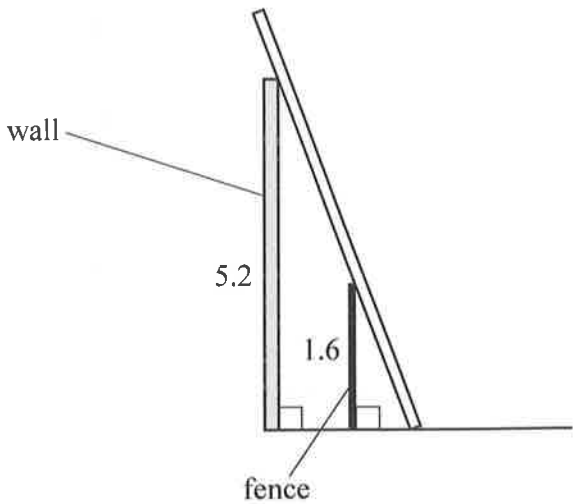
(a) Show that the safe angle between the ladder and the horizontal is 76°, correct to the nearest degree.

Answer

[2]

(b) Lihui uses the 4 to 1 rule to place a ladder safely against a wall, as shown in the diagram.

The wall is vertical and 5.2 m high.
A fence of height 1.6 m is parallel to the wall.
The ladder just touches the top of the fence.



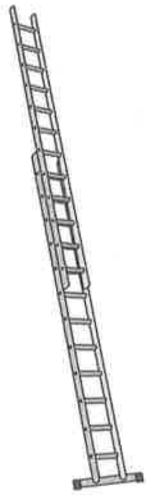
(i) Calculate the distance of the bottom of the ladder from the bottom of the wall.

Answer .... m [1]

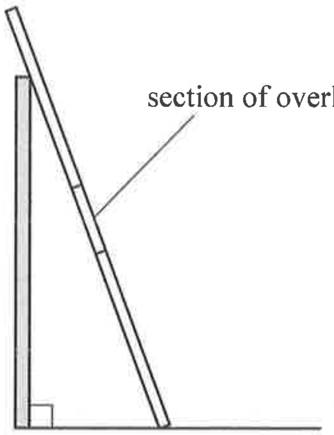
(ii) Calculate the distance of the bottom of the fence from the bottom of the wall.

Answer .... m [2]

(c) Huang has a 2-section ladder.
The length of each section is 3.4 m.



Huang places the ladder against a wall that is 4.7 m high.
The ladder extends 1 m beyond the top of the wall.
The sections of the ladder overlap by 0.9 m.



(i) Calculate the angle between the ladder and the horizontal ground.

Answer .... [2]

(ii) Is the ladder placed in a safe position?
Give a reason for your answer.

Answer .... [1]

Section B (8 marks)

Answer **one** question from this section. Each question carries 8 marks.

11 **(a)** The table shows the lengths of time, t minutes, that 150 students from school A took to do a task.

Time, t minutes	Frequency
$10 \leqslant t \leqslant 20$	13
$20 < t \leqslant 25$	27
$25 < t \leqslant 30$	44
$30 < t \leqslant 35$	38
$35 < t \leqslant 40$	19
$40 < t \leqslant 50$	9

(i) Tony says that the range of the times is 40 minutes.

Give a reason why Tony may be wrong.

Answer ....

.... [1]

(ii) Calculate estimates for the mean and standard deviation of the times.

Answer Mean .... minutes

Standard deviation .... minutes [2]

(iii) The mean time taken by students at school B to do the same task is 33.2 minutes.
The standard deviation of the times for school B is 6.2 minutes.

Which school has less variation in times?
Give a reason for your answer.

Answer School .... because ....

.... [1]

(b) There are 5 white balls and 3 black balls in a bag.
Hadil picks 2 balls at random from the bag.

Find the probability that

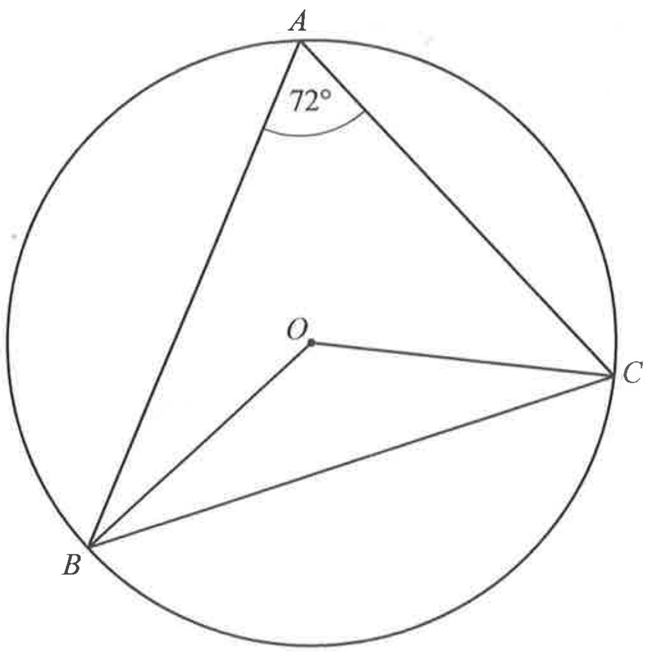
(i) both balls are white,

Answer .... [2]

(ii) the balls are of different colours.

Answer .... [2]

12 (a)



A, B and C are points on the circle, centre O.
Angle $BAC = 72°$.

Find angle OCB.
Give reasons for your answer.

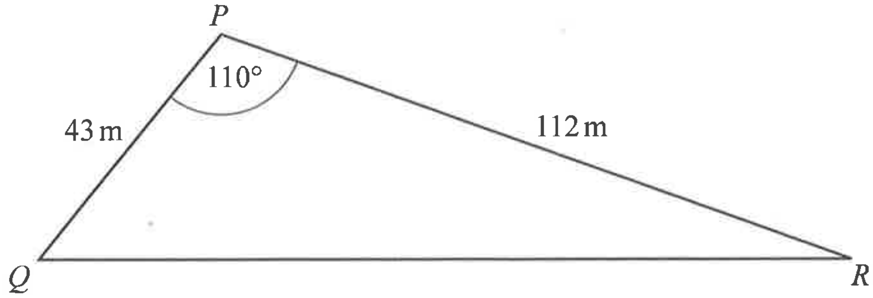
Answer Angle OCB = ... Reasons ....

....

....

[3]

(b)

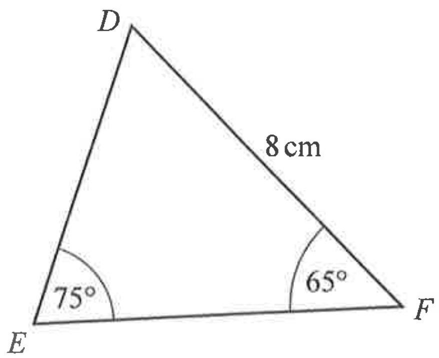


$PQ = 43$ m and $PR = 112$ m.
Angle $QPR = 110°$.

Calculate QR.

Answer $QR =$ .... m [2]

(c)



$DF = 8$ cm.
Angle $DEF = 75°$ and angle $DFE = 65°$.

Calculate EF.

Answer $EF =$ .... cm [3]

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